

Lecture VIII - Pandas and AI Craft

Programming with Python

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Checkpoint 4

Sessions VI-VII. The first 40 minutes are the checkpoint. It starts now, before the investor opens her data room.

- **Individual work:** no neighbors, no chat
- **AI tools are allowed:** being able to **VERIFY** the output is the skill being graded
- The **link is on Moodle**. Open it and start
- ~6 short tasks: write code, fix a bug, a seeded draw, answer a multiple choice
- It sweeps **Sessions VI-VII:** modules, `random` and seeds, NumPy arrays and masks

When you're done

Menu → *Download* → *Download Python code* → upload the `.py` to the “**Checkpoint 4**” assignment on Moodle. **No retakes:** one sitting.

...

Note

The green live checks are **provisional**. The final grading runs on my side. Everything in it was rehearsed in the labs.

Episode 8: The Data Room

Show me the data

Pens down. The checkpoint is behind you. This morning the rival chain MunchCorp put out a press release bragging about their “data-driven growth.” The evidence attached: exactly **one pie chart**. The investor slid it across the table and said, “*Yours will be better. Show me the data.*”

...

Tobi didn't wait. He'd pasted the question into a chatbot, proudly turned the laptop around, and hit run:

```
AttributeError: 'DataFrame' object has no attribute 'overview'
```

...

The AI sounded certain. The method it called **does not exist**. Today is about that difference, and about the tool that answers the investor for real: **pandas**.

AI joins the team: professionally

Prompting: context + constraints

Two things turn a vague request into a useful answer:

- **Context.** Say what the data is: “I have a DataFrame `orders` with columns `zone` (text) and `total_eur` (float).”
- **Constraints.** Say exactly what you want *back*: “Give me the total `total_eur` for zone `Nord`, as a single number.”

...

“DataFrame `orders`, text column `zone`, float column `total_eur`. Write one line that returns the total `total_eur` where `zone` is `Nord`.”

Vague in, vague out. Specific in, checkable out, and then you **iterate**.

The verify workflow

Tobi’s mistake wasn’t *using* AI. It was **shipping without checking**. Every line an AI hands you gets three passes:

1. **Read it:** do you understand what each line claims to do?
2. **Run it:** does it actually execute, or does it crash?
3. **Test it:** try it on a small case where **you already know the answer**.

Step 3 in action

Five orders with values you can add in your head, and the AI’s suggested line for the Nord average:

```
import pandas as pd

check = pd.DataFrame({
    "zone":      ["Nord", "Sued", "Nord", "Hafen", "Nord"],
    "total_eur": [10.0, 7.5, 20.0, 5.0, 30.0],
})
print(check[check["zone"] == "Nord"]["total_eur"].mean())
```

```
20.0
```

...

The Nord orders are 10, 20 and 30. You can average those in your head. Does the AI’s answer match? Then the line has earned some trust on eighty rows. Verification is the job now, not typing.

Predict: it ran, so it's right?

Tobi asks the AI for the **Nord average** on the same five orders. It hands back one line and calls it done. What does this print?

```
print(check["total_eur"].mean())
```

a) 20.0 b) an `AttributeError` c) 14.5

...

Predict. Pick a letter, then I reveal the answer.

Answer: it ran, and it's wrong

```
print(check["total_eur"].mean())           # every zone
print(check[check["zone"] == "Nord"]["total_eur"].mean()) # Nord only
```

```
14.5
20.0
```

...

c) **14.5**: the line averages **all five** orders. Nobody filtered for Nord, and pandas has no way to know that was the question.

...

No traceback, a plausible number, and still wrong. "It runs" only clears step 2; step 3 is where the investor's number gets tested.

Predict: does `.summarize()` exist?

The AI wrote this for Tobi. `orders` is a tiny two-row frame. What happens on the last line?

```
import pandas as pd
orders = pd.DataFrame({"zone": ["Nord", "Sued"], "total_eur": [12.0, 9.5]})

orders.summarize()
```

a) prints a summary table b) raises an `AttributeError` c) returns an empty DataFrame

...

Predict. Pick a letter, then I reveal the answer.

Answer: the method never existed

```
import pandas as pd
orders = pd.DataFrame({"zone": ["Nord", "Sued"], "total_eur": [12.0, 9.5]})
orders.summarize()           # the method the AI invented
```

```
AttributeError: 'DataFrame' object has no attribute 'summarize'
[31mAttributeError[39m[31m: [39m 'DataFrame' object has no attribute
'summarize'
```

...

b) **AttributeError**: pandas has no `.summarize()`. The AI invented it.

...

The real method is `.describe()`:

```
print(orders.describe())          # the method that actually exists
```

```
      total_eur
count    2.000000
mean    10.750000
std      1.767767
min      9.500000
25%     10.125000
50%     10.750000
75%     11.375000
max     12.000000
```

...

An AI that sounds sure is not the same as an API that exists.

Why did it invent a method?

Session VI gave you the machine, so here is why it fails in exactly this way.

- Training rewards a plausible continuation and never checks a true one: nothing in the loop verifies facts
- And it always writes *some* next token, confident or not: a choice gets picked, printed, and it all looks equally smooth
- `.summarize` is a *plausible* pandas name: short, lowercase, shaped like `.describe()`. Plausible is exactly what the machine optimizes for

...

Fluent and wrong is not a bug to patch. It is how the machine works.

How it was trained, in one slide

- **Pre-training**: predict the next token over massive text, where the weights inside get adjusted; applying them afterwards is cheap
- **Fine-tuning**: curated examples teach it to follow instructions
- **RLHF** (reinforcement learning from human feedback): humans rank answers; it learns to prefer the ranked-higher ones

- Its knowledge has an end date, the knowledge cutoff: after it, it guesses fluently instead. And it inherits the biases of the text it read

...

Every stage rewards *sounding* right; none of them checks pandas. That is your job, and step 3 of the verify workflow is how.

Disclosure: one line, every time

From Session VI the course rule stands: every submission that used AI carries a **one-line note** saying what you used it for.

- “Used the chatbot to draft the `groupby` line; I checked the totals by hand.”
- Standard practice, not an admission of anything.

...

It protects **you**: it separates the work you understand from the work you pasted, so when a reviewer (or the investor) asks “how does this line work?”, you’re never caught claiming something you can’t explain.

When NOT to reach for AI

Sometimes the fastest path is the one you already own. You built a whole muscle in Part I:

- A red traceback? **You’ve read those since Episode 5.** The last line names the error and points at the file, often faster than describing it to a chatbot.
- A one-line filter you’ve written ten times? Just write it.

...

Tip

Use AI to draft the unfamiliar and to explain the confusing, not to skip thinking you can do yourself.

Your turn: 5-10 minutes

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_08_a/



Any questions? Let me know!

The DataFrame

What pandas adds over NumPy

Last session, a NumPy array turned a thousand numbers into one fast object, but every value had to share **one type**, and columns had no names.

- A **DataFrame** is a table: named columns, and each column can be a different type
- Text zones next to float euros next to integer counts, all in one object
- It's the spreadsheet the investor handed you, loaded into Python

...

The convention everyone uses: `import pandas as pd`.

A DataFrame from a dictionary

Keys become **column names**; each list becomes a **column**; one order per row, mixed types together:

```
import pandas as pd
df = pd.DataFrame({
    "order_id": [101, 102, 103, 104, 105],
    "zone":     ["Nord", "Sued", "Nord", "Hafen", "Sued"],
    "items":    [2, 1, 3, 1, 4],
    "total_eur": [18.50, 7.20, 24.00, 6.80, 31.40],
})
print(df)
```

	order_id	zone	items	total_eur
0	101	Nord	2	18.5
1	102	Sued	1	7.2

2	103	Nord	3	24.0
3	104	Hafen	1	6.8
4	105	Sued	4	31.4

...

Note

Nobody types eighty orders by hand: today's lab loads a real CSV in one line, `orders = pd.read_csv("public/orders.csv")`. In the browser that line fetches over the web instead of from disk; your code doesn't change.

First look: `.head()` and `.info()`

Before analyzing a table, glance at it. `.head()` shows the top rows; `.info()` is your first sanity check: right number of rows? Any column a surprising type?

```
print(df.head(3))    # first 3 rows
df.info()            # columns, dtypes, non-null counts
```

```
   order_id  zone  items  total_eur
0        101  Nord     2        18.5
1        102  Sued     1         7.2
2        103  Nord     3        24.0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   order_id    5 non-null        int64
1   zone        5 non-null        object
2   items       5 non-null        int64
3   total_eur   5 non-null        float64
dtypes: float64(1), int64(2), object(1)
memory usage: 292.0+ bytes
```

The numbers at a glance: `.describe()`

`.describe()` computes count, mean, min, max and quartiles for every **numeric** column at once:

```
print(df.describe())
```

	order_id	items	total_eur
count	5.000000	5.000000	5.000000
mean	103.000000	2.200000	17.580000

std	1.581139	1.30384	10.688873
min	101.000000	1.00000	6.800000
25%	102.000000	1.00000	7.200000
50%	103.000000	2.00000	18.500000
75%	104.000000	3.00000	24.000000
max	105.000000	4.00000	31.400000

...

One call, the whole numeric summary: this is the `.summarize()` the AI wished existed, spelled correctly.

Selecting a column, filtering rows

Pick one column by name; keep rows with a **boolean mask**, exactly the NumPy idea from last session:

```
print(df["zone"])          # one column (a Series)
print()
print(df[df["zone"] == "Nord"])  # only the Nord rows, case-sensitive!
```

```
0    Nord
1    Sued
2    Nord
3    Hafen
4    Sued
Name: zone, dtype: object
```

	order_id	zone	items	total_eur
0	101	Nord	2	18.5
2	103	Nord	3	24.0

...

`df["zone"] == "Nord"` builds a column of `True/False`; indexing with it keeps the `True` rows. `.loc` exists for label-based selection, but a plain mask covers today.

Predict: what does a comparison return?

Tobi wants the orders over 20 EUR. He compares the column and prints the result:

```
big = df["total_eur"] > 20
print(big)
```

a) a column of `True/False`, one per row b) only the rows whose total is over 20 c) a single `True`, since some order is over 20

...

Predict. Pick a letter, then I reveal the answer.

Answer: a mask, not a table

```
big = df["total_eur"] > 20
print(big)           # the mask: a Series of booleans
print(df[big])       # the rows: build the mask, then apply it
```

```
0    False
1    False
2     True
3    False
4     True
Name: total_eur, dtype: bool
   order_id  zone  items  total_eur
2        103  Nord     3         24.0
4        105  Sued     4         31.4
```

...

a) a column of **True/False**. The comparison alone is the **mask**, a Series with one boolean per row. The rows only appear once you index with it.

Your turn: 5-10 minutes

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_08_b/



Any questions? Let me know!

Adding a column, sorting

Compute a new column from existing ones, then sort. Work on a **copy** so the original stays intact:

```
priced = df.copy()
priced["eur_per_item"] = priced["total_eur"] / priced["items"]
print(priced.sort_values("total_eur", ascending=False))
```

	order_id	zone	items	total_eur	eur_per_item
4	105	Sued	4	31.4	7.85
2	103	Nord	3	24.0	8.00
0	101	Nord	2	18.5	9.25
1	102	Sued	1	7.2	7.20
3	104	Hafen	1	6.8	6.80

...

`sort_values` reorders rows by a column, biggest orders first here. The new column is just arithmetic on two others, computed for every row at once.

Predict: filtering with the wrong case

Tobi filters for the Nord zone, but types it **lowercase**. The data spells it "Nord". What does this print?

```
print(df[df["zone"] == "nord"])
```

a) an empty table, no error b) the Nord rows anyway: case is ignored c) a `KeyError` for the missing zone

...

Predict. Pick a letter, then I reveal the answer.

Answer: an empty table, no warning

```
print(df[df["zone"] == "nord"])          # nothing matches "nord"
print("rows:", len(df[df["zone"] == "nord"]))
```

```
Empty DataFrame
Columns: [order_id, zone, items, total_eur]
Index: []
rows: 0
```

...

a) an empty table. "nord" matches nothing, so the mask is all `False` and pandas hands back **zero rows**. No error, no complaint.

...

This is the dangerous kind of bug: it **runs**, it just returns nothing. **pandas won't warn you; YOU verify**, which is exactly why you test filters on a case whose answer you know.

One more tool: `groupby`

The investor's real question is **per zone**: which area brings in the most?

- `groupby` splits the table by a category, then computes **once per group**
- `df.groupby("zone")["total_eur"].sum()` → one total for each zone

...

Note

That's the teaser: today's lab does the heavy lifting with `groupby`, turning eighty raw orders into the handful of numbers the investor actually asked for.

Your turn: 5-10 minutes

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_08_c/



Any questions? Let me know!

To the Lab

After the break: the lab

- Head to the lab notebook: [Episode 8: The Data Room](#), in your browser as always
- The investor slides a **USB stick** across the table: every order of two weeks, now `orders.csv`, eighty rows, one `pd.read_csv` line
- You'll `.head()` it, ask its `.shape`, filter with masks, add a column on a safe copy, and answer per-zone questions with `groupby`

- Tobi has discovered AI; your real job is to **supervise** it and catch the confident nonsense

...

! Important

Download your .py before you leave. Closing the tab without downloading loses your work, the same motion you used to hand in the checkpoint at the start of the session.

Wrap-up

Three things to remember

1. **AI drafts, you verify.** Give it context and constraints, then read it → run it → test it on a case you know.
2. **A DataFrame is a named, mixed-type table.** `pd.DataFrame` from a dict, `pd.read_csv` from a file; `.head()`, `.info()`, `.describe()` to look before you leap.
3. **Select, filter, add, group.** `df["col"]`, a boolean mask (`df[df["zone"] == "Nord"]`, case-sensitive!), a new column on a `.copy()`, and `groupby` for per-category answers. pandas won't warn you; you verify.

...

i Note

Next episode: charts the investor can't argue with. Numbers convince the careful; a good plot convinces the room. We turn the data room into pictures.

Literature

Books to start with

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. [Link to free online version](#)
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

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i Note

Working with AI this session? Revisit the [AI Tools page](#): context-and-constraints prompting, the verify workflow, and the one-line disclosure habit. For pandas itself, the official [10 minutes to pandas](#) guide is the friendliest next step.

...

For more, see the [literature list](#) of this course.